

ClassIQ - Acceptance Test Plan

ClassIQ - Teacher Grade Book and Report Card System

Purpose: define measurable acceptance criteria and design acceptance tests for the ClassIQ system

References and Basis

This plan is based on the inspected ClassIQ codebase and the Sprint requirement document. The current implementation shows teacher and student login flows, multilingual UI support (English, Sinhala, Arabic), RTL support for Arabic, localized database queries, teacher marksheet management, student details and feedback handling, student grade viewing, and report card/PDF generation.

The plan covers functional testing, usability testing, and performance and reliability testing.

Tests are designed as end-to-end business scenarios where possible, with observable pass or fail outcomes.

Product Scope Summary

Teacher-side capabilities:

login, view student information, enter or edit marks by subject, calculate totals and grades, clear marks, review student details, and save or delete feedback.

Student-side capabilities:

login, view personal information, view grading criteria, view marks and report card, and download or export report card.

Cross-cutting capabilities:

localization in English, Sinhala, and Arabic, database-level translation fallback, and right-to-left layout for Arabic.

Defined Acceptance Criteria

1. User authentication

Teachers and students can log in with valid credentials and are rejected with clear errors for invalid or incomplete credentials

2. Role-based access

Teacher users reach teacher dashboard features; student users reach student dashboard features; role mismatch is blocked

3. Teacher marks management

Teachers can enter, update, and clear marks for their subject, and saved values persisted correctly in the database.

4. Grade calculation

Subject totals and grades are calculated consistently from entered components and displayed correctly to users.

5. Student information access

Teacher can retrieve student details and student can view own information without editing capability.

6. Feedback management

Teacher can save, retrieve, and delete feedback for a selected student, and students see the latest feedback on the report card view.

7. Report card generation

Student report card view shows subjects, total, average, and feedback; PDF export completes successfully.

8. UI localization

UI text changes correctly among English, Sinhala, and Arabic without restart, using resource bundles.

9. Database localization

Localized values retrieved from translation tables are shown in the selected language with English or base fallback when a translation is missing.

10. RTL support

Arabic screens use right-to-left orientation for navigation and content alignment where applicable.

11. Usability quality

Key tasks such as login, enter marks, view grades, and export report card can be completed with clear labels, readable layouts, and understandable feedback messages.

12. Performance and reliability

Common user actions complete within agreed response targets, and the application remains stable under expected classroom-size usage and recoverable failures.

Coverage Matrix

Number	Description	Mapped criteria	Mapped test cases
01	Teacher and student login functionality	1 & 2	F-1, F-2, F-3
02	Teacher manages marks / grade entry update	3	F-4, F-5, R-2
03	Automatic calculation of averages and grades	4	F-6, F-7
04	Teacher provides feedback	6	F-8, F-9
05	Students view marks, grades, feedback	5 & 6	F-10, F-11
06	Generate report cards / download report cards	7	F-12, F-13
07	Multilingual UI and dynamic language switching	8	F-14, U-3
08	Localized database content support	9	F-15, F-16
09	RTL layout support for Arabic	10	U-4
10	Readable, efficient and error-minimizing user experience	11	U-1, U-2
11	Stable performance and recovery	12	P-1, P-2, R-1, R-3

Example Acceptance Test Cases

Functional Testing

- F-1 - Valid teacher login
- F-2 - Valid student login
- F-3 - Reject invalid credentials
- F-4 - Save subject marks
- F-5 - Clear marks for teacher subject
- F-6 - Validate total and grade display
- F-7 - Student report card totals
- F-8 - Save feedback
- F-9 - Delete feedback
- F-10 - Teacher views localized student details
- F-11 - Student views own information only
- F-12 - Open report card page
- F-13 - Export report card PDF
- F-14 - Dynamic UI language switching
- F-15 - Database translation retrieval
- F-16 - Fallback when translation missing

Usability Testing

- U-1 - Teacher completes marks entry task efficiently
- U-2 - Student finds grades and report card
- U-3 - Language selection clarity
- U-4 - Arabic RTL readability

Performance and Reliability Testing

- P-1 - Page response time
- P-2 - PDF generation time

R-1 - Repeated save reliability

R-2 - Input validation reliability

R-3 - Database interruption handling

Conclusion

The purpose of this document is to confirm that all major functional and non-functional requirements have been translated into clear acceptance criteria and measurable test scenarios. The coverage matrix demonstrates that each important requirement is linked to one or more acceptance tests. These tests are intended to support final system review and user acceptance before project completion.